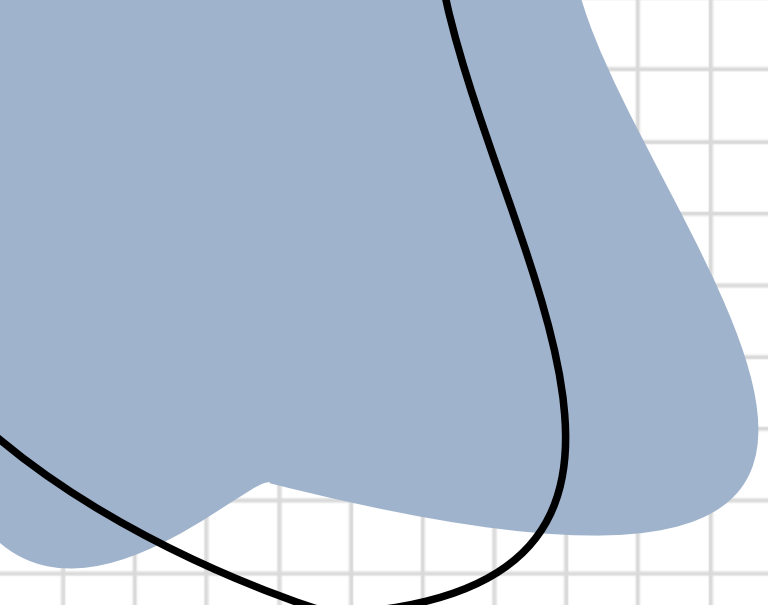
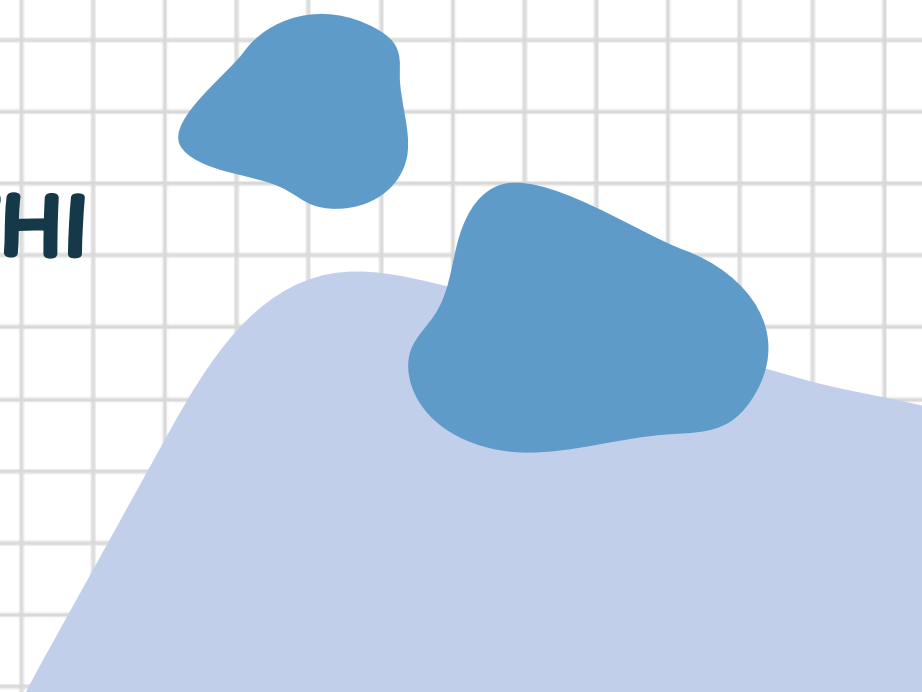




Agentic AI for Dynamic EV Charging Tariff Optimization

Demand Forecasting, Dynamic Pricing and Continuous Learning using ACN-Data and UrbanEV

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23125012



DATA LANDSCAPE & PREPROCESSING

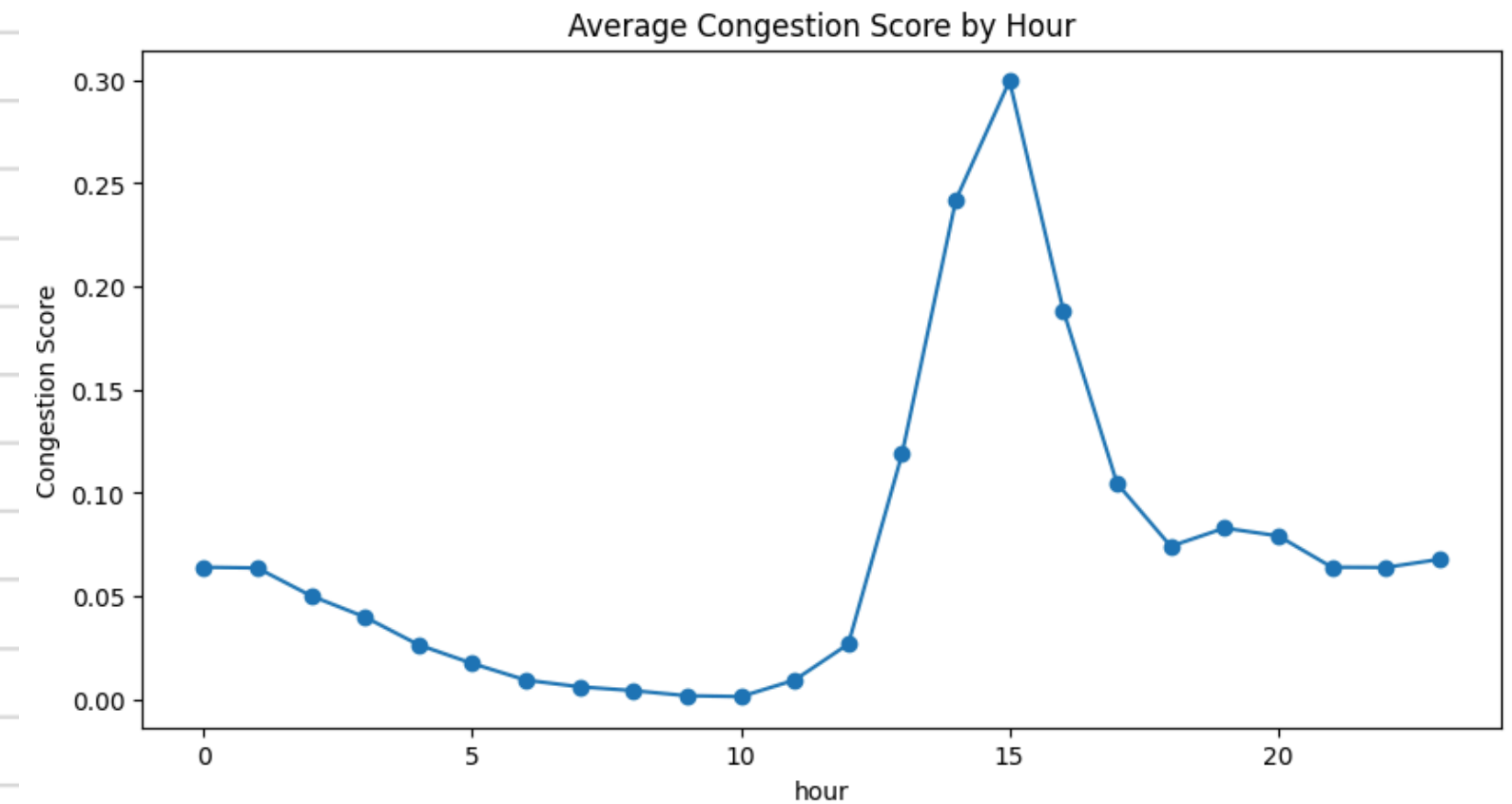
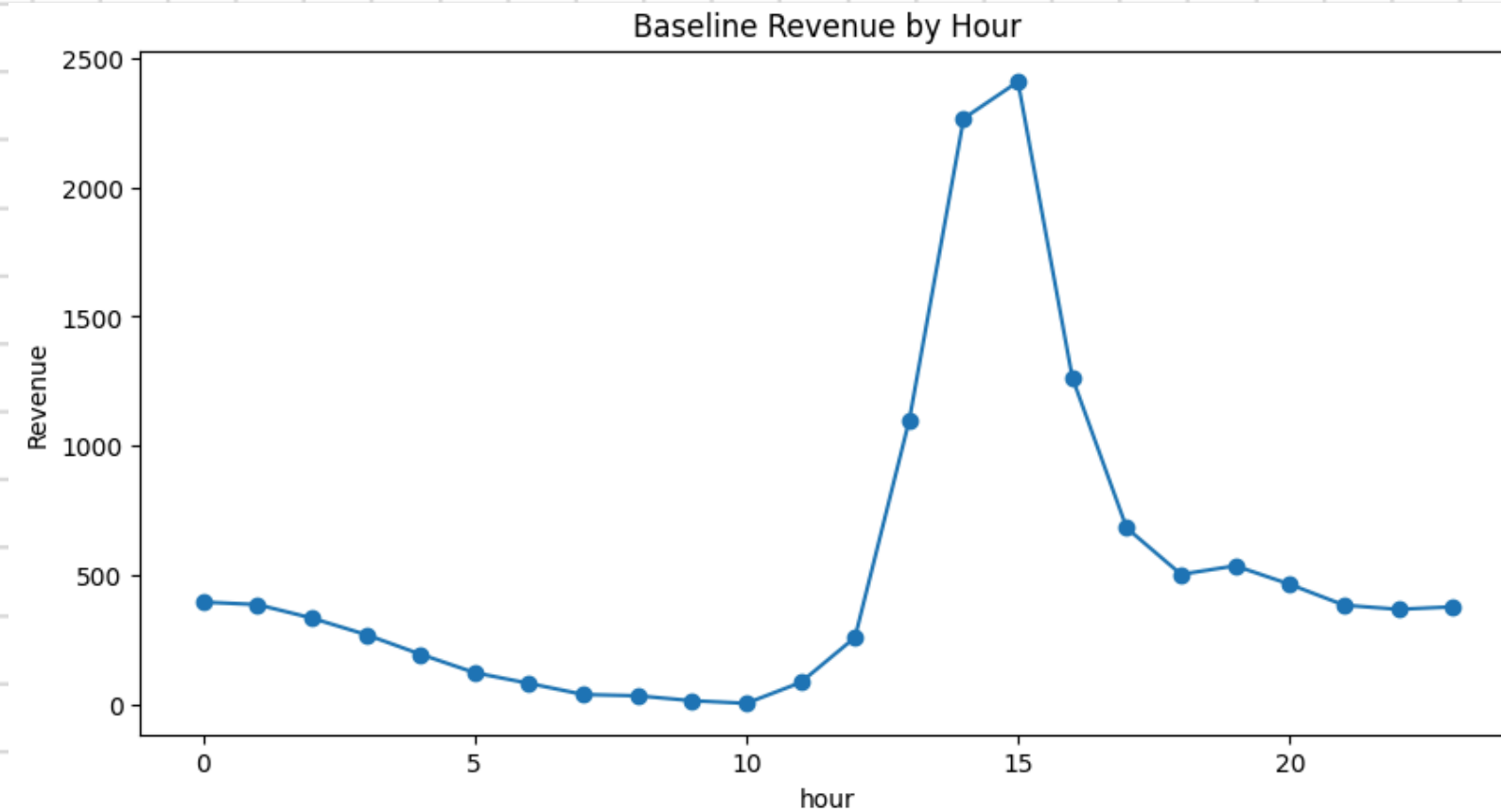
Datasets used:

Dataset	Description	Records
ACN-Data (Caltech + JPL)	EV charging sessions, energy delivered, user behavior	31,900+ sessions
UrbanEV	Occupancy, charging volume, duration, pricing	8,640 timestamps × 247 stations

Preprocessing Steps:

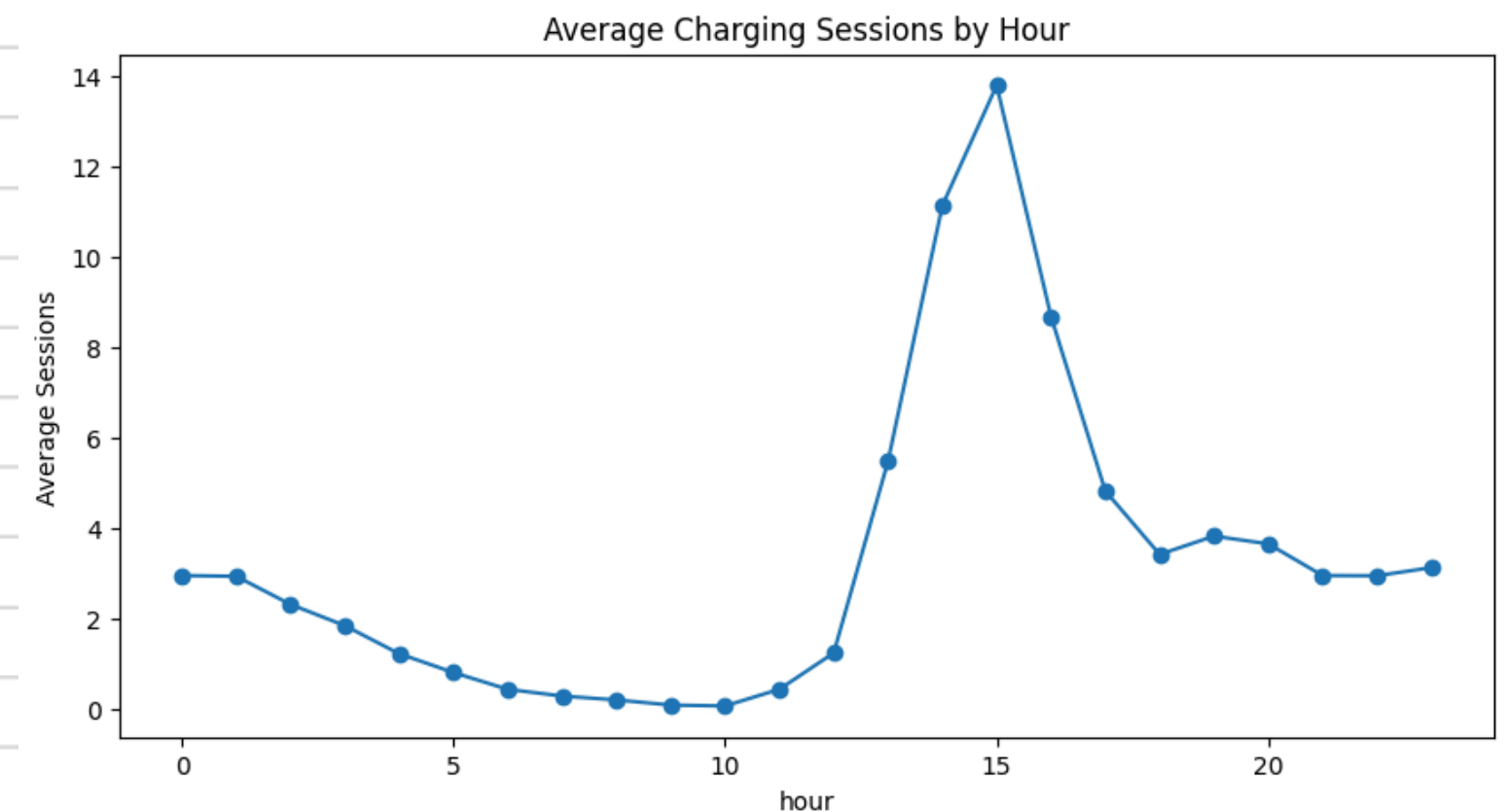
- ✓ Merged Caltech and JPL datasets
- ✓ Converted timestamps to datetime format
- ✓ Created charging duration and utilization features
- ✓ Estimated baseline revenue
- ✓ Generated hourly demand dataset
- ✓ Created lag and rolling features for forecasting

EDA & DEMAND BEHAVIOR



Observed:

- High congestion during afternoon
- Revenue concentrated in peak hours
- Significant demand fluctuations



DEMAND PREDICTION AGENT

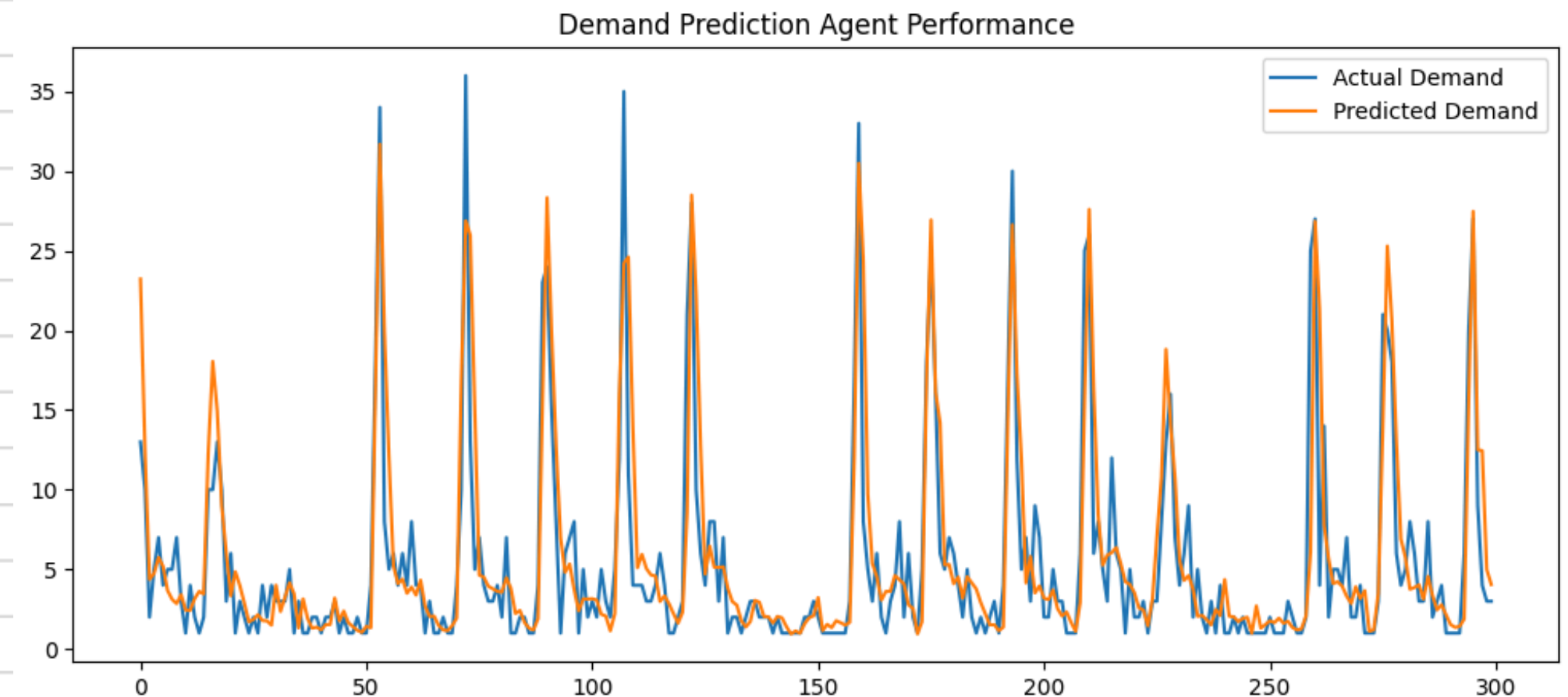
Objective: Predict future EV charging demand

Features Used:

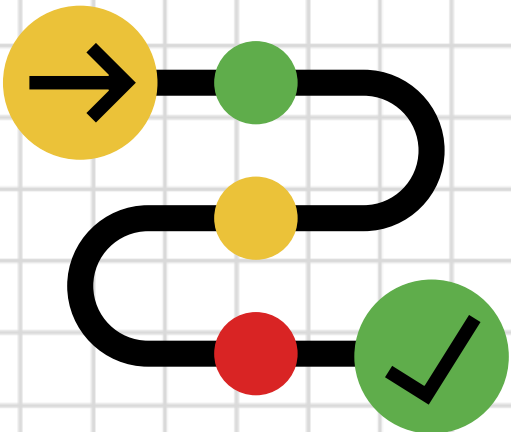
- Lag features
- Rolling averages
- Time features
- Cyclic features
- Historical energy demand

Model Comparison:

MODEL	R2
Random forest	0.447
XGBoost	0.738



DYNAMIC TARIFF PRICING AGENT



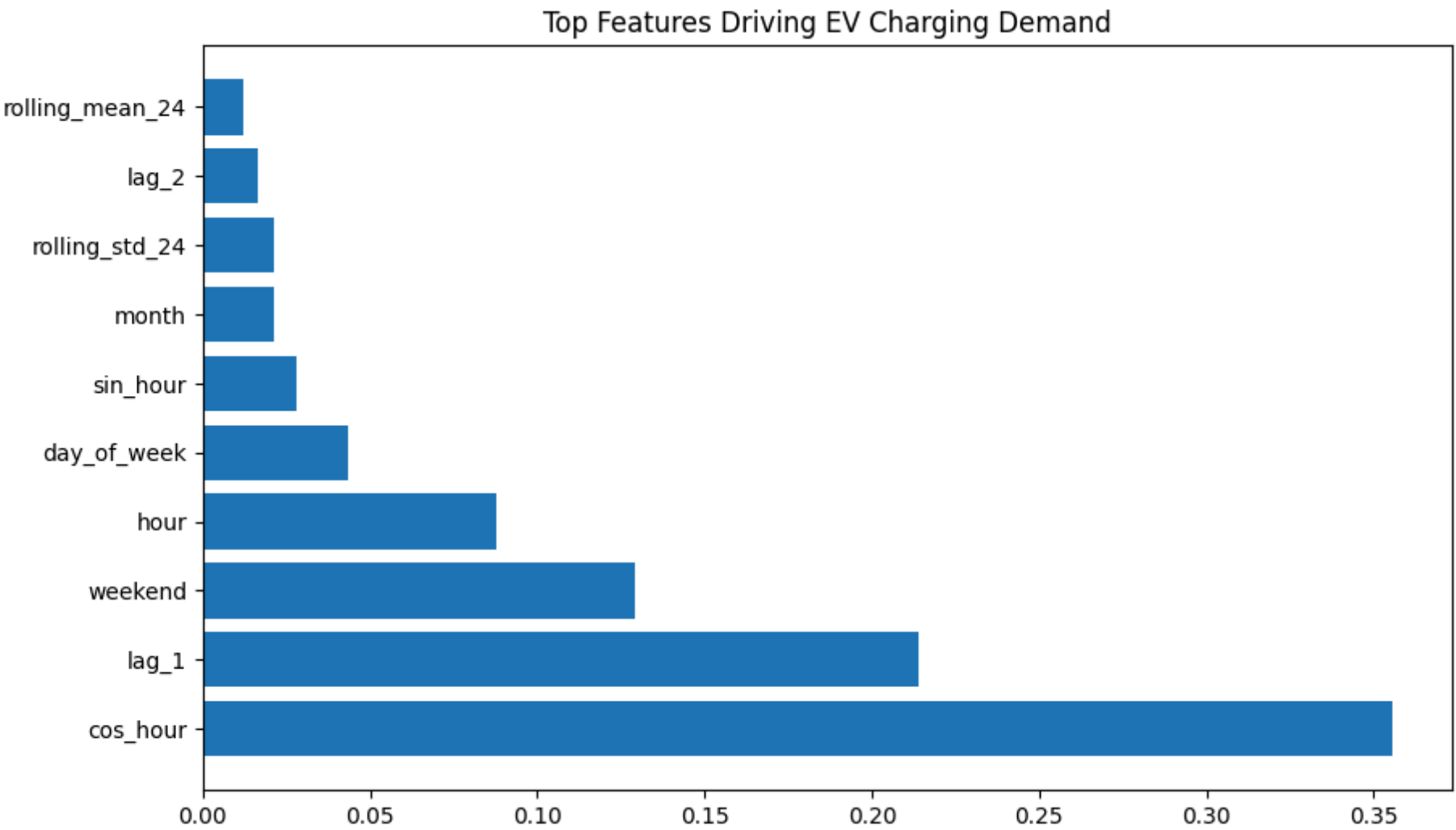
Pricing Strategies Evaluated:

STRATEGY	REVENUE GAIN
Rule based	5.28%
Aggressive	5.63
Percentile based	22.95%

Selected Strategy: Percentile-Based Dynamic Pricing

Logic:

Top 25% demand → Surge Pricing
Middle Demand → Standard Pricing
Lowest Demand → Discount Pricing



MONITORING & LEARNING AGENT

Agent Responsibilities:

Monitor:Revenue,Utilization,Congestion,Pricing efficiency

Results:

METRIC	VALUE
Revenue Gain	22.95%
Pricing Efficiency	18.44
Avg Utilization	70.17%

Feedback Logic:

Revenue Gain >15%
→ Keep Policy

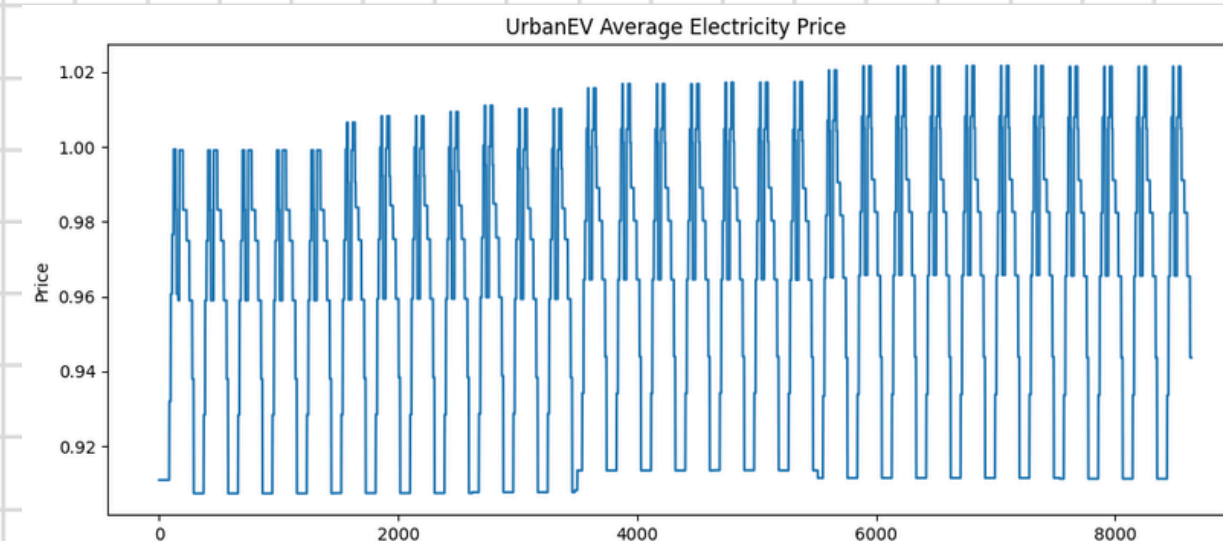
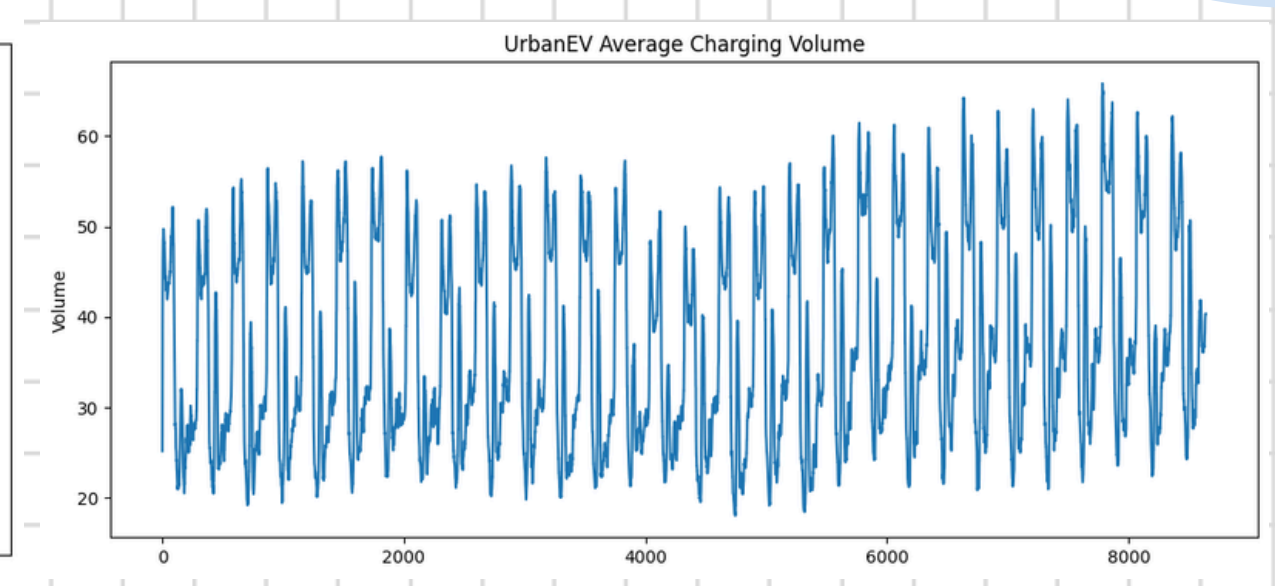
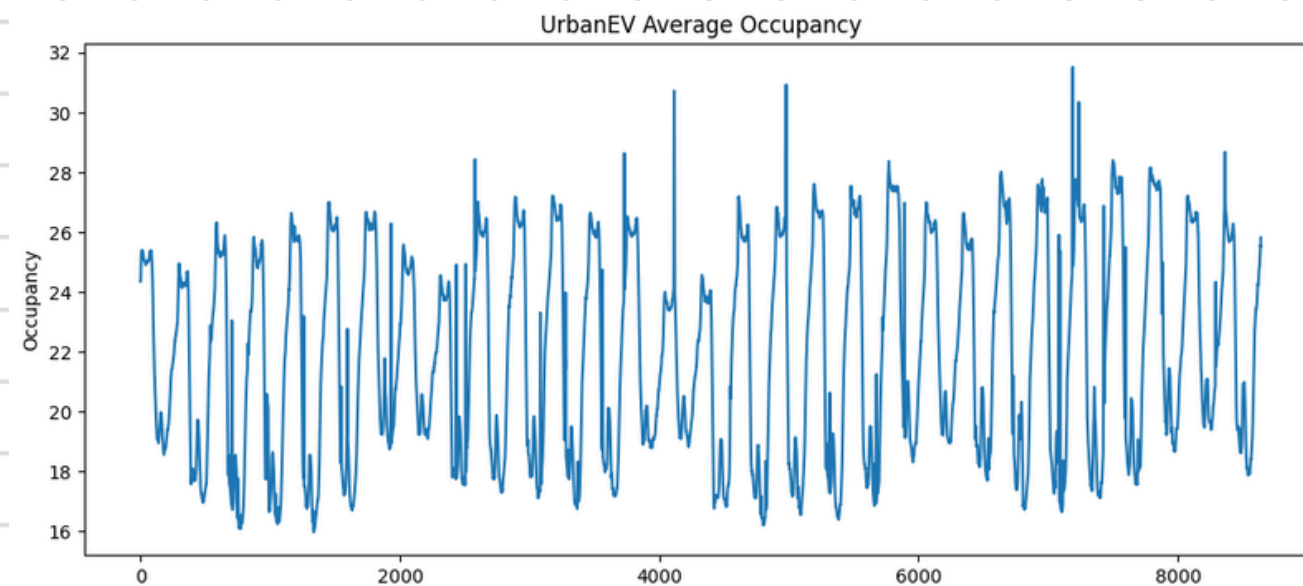
Revenue Gain 5-15%
→ Minor Adjustment

Revenue Gain <5%
→ Retrain Pricing Agent

URBANEV VALIDATION & ROBUSTNESS

Observations:

Metric	Value
Avg Occupancy	21.9
Avg Volume	36.68
Avg Duration	1.47
Avg Price	0.96



Conclusion:

UrbanEV demonstrates:

- Demand variability
- Occupancy fluctuations
- Price diversity

These support adaptive pricing strategies.

BUSINESS IMPACT & POLICY IMPLICATIONS

Business Impact: Revenue increased 22.95% without infrastructure expansion

Operational Benefits:

- Better charger utilization
- Reduced congestion risk
- Demand-aware charging ecosystem

Policy Implications:

- Supports Time-of-Use pricing
- Encourages off-peak charging
- Improves grid stability

The proposed Agentic AI framework successfully forecasts EV demand, optimizes charging tariffs, and continuously adapts pricing policies, resulting in a 22.95% increase in revenue while maintaining healthy charger utilization.

Thank you